Election Logistics

IE301 - Group 8 - Presentation 2

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FACTORS USED TO EVALUATE CONSTITUENCIES

Goal: Divide the region into (k) nonempty constituencies, assigning every geographic unit exactly once.

POPULATION BALANCE

min $\Sigma (P_j - \bar{P})^2$

Target $\bar{P}$ = region population ÷ k.

Minimise the sum of squared deviations from that target across all k districts.

COMPACTNESS

min perimeter^2 / area

Lower values mean tighter shapes.

Lower values favour compact shapes and discourage sprawling, irregular boundaries.

ACCESSIBILITY

min mean travel time

Population-weighted road time.

Weight by population, and check the maximum too, so a good average does not hide remote settlements.

HARD CONSTRAINTS

- Every constituency is one connected region
- Population stays inside the permitted band
- Every geographic unit assigned exactly once

Selection rule: Compare maps using a weighted combination of population imbalance, compactness and accessibility penalties. Higher scores indicate better performance under the chosen priorities.

OFFICIAL INDIAN DELIMITATION HEURISTIC

1. State Target Quota ($\bar{P}$State)
Total State Pop ÷ Total Assembly Seats

↓

2. Parliamentary Multiplier (M)
Total Assembly Seats ÷ Total Lok Sabha Seats

↓

3. District Seat Entitlement (Kd)
District Pop ÷ $\bar{P}$State ⇒ Round to Whole Integer

↓

4. District Quota & Permissible Band
$\bar{P}$d = District pop ÷ Kd ⇒ Band: [0.90$\bar{P}$d, 1.10$\bar{P}$d]

↓

5. Serpentine / Zigzag Ground Sweep
Start North-East → Group unbroken villages → Seal AC

↓

6. Constitutional SC/ST Reservation
ST: Largest % in descending order | SC: Dispersed across state

↓

7. National Aggregation (Lok Sabha PCs)
Bundle M contiguous Assembly Constituencies into 1 MP Seat

↓

8. Public Sittings & Final Gazette Order
Final order published as non-justiciable statutory law

Key Characteristics

- Legal Basis: Delimitation Act, 2002; ECI Guidelines (2004).
- Hierarchical Invariance: Revenue villages/wards are indivisible atomic units.
- Limitation: Single-pass manual sweep; cannot simultaneously optimize shape compactness or citizen travel distances.

GENETIC ALGORITHM OPTIMIZATION

1. Generate Draft Maps
Create 100 random contiguous graph partitions

↓

2. Multi-Objective Fitness Evaluation
Grade: (1) Pop Balance + (2) Compactness + (3) Logistics Cost

↓

3. Selection
Keep high-scoring parent maps, discard poor ones

↓

4. Spatial Crossover
Combine best regional subzones from parent maps

↓

5. Boundary Mutation & Repair
Shift border villages to balance populations; fix disconnected enclaves

↓

Stopping Criteria?
(e.g., 500 gens / converged)

NO → (back to 2. Multi-Objective Fitness Evaluation)

YES ↓

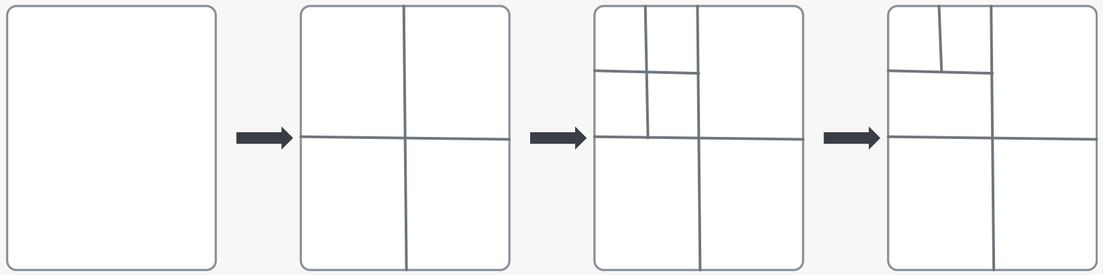
Final Delimitation Map
Globally optimized, compact, and accessible

THE QUADTREE APPROACH

A recursive spatial partitioning approach that divides a region into four smaller areas until population thresholds are satisfied, followed by merging of adjacent under-populated regions.

WHAT WE WOULD ADD

- Apply it to a real map
- Add minimum as well as maximum population thresholds
- Add compactness and contiguity, which the method ignores
- Validate by sampling, to check for orientation bias



split → split → split → merge

PSEUDOCODE

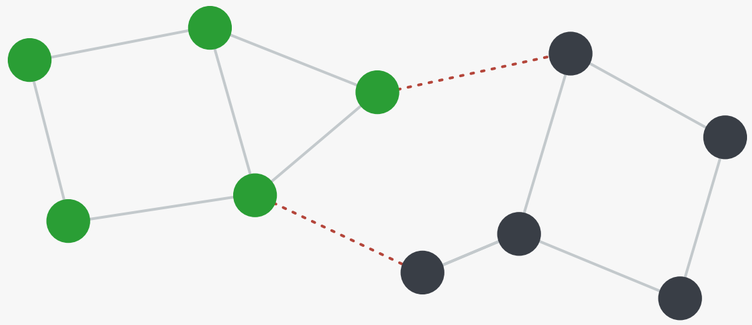
Input: region, population data, threshold Th

1. Make the whole region the root node
2. For each region:
   calculate its population p
   if p > Th: split into 4 quadrants,
   then recurse on each
   else: mark it a leaf node
3. For each pair of adjacent leaves:
   if same parent and combined p ⩽ Th,
   merge the two nodes
4. Repeat merging while any pair qualifies
5. Output the regions as constituencies

Reference: arxiv.org/pdf/2402.09336

THE PROBLEM AS A BALANCED GRAPH PARTITIONING

Core idea – Represent the region as a graph and divide its nodes into a fixed number of groups. Each group should have approximately equal total node weight while keeping strongly connected nodes together



NODES	One per village, weighted by its population.
EDGES	Connect villages that share a geographic boundary.
EDGE WEIGHTS	Connection strength. Nearby villages stronger, distant villages weaker.

Three Heuristics for Graph Partitioning

1. Kernighan–Lin Algorithm

Start with existing groups. Evaluate exchanges of nodes between them and apply sequences of swaps that reduce the total weight of crossing edges. Repeat until further exchanges provide no improvement.

Key feature: Improves an existing partition through local adjustments. Standard swaps preserve node counts, so population-weighted balance requires additional checks.

2. Multilevel Graph Partitioning

Merge neighbouring nodes into larger units, combining their population weights. Partition the resulting smaller graph, then gradually restore the original nodes and refine their assignments while maintaining balance.

Key feature: Handles large graphs efficiently by making broad grouping decisions first and refining smaller details afterward.

3. Weighted K-Means Clustering

Initialize cluster centres using k-means. Assign each node to the cluster with the lowest weighted distance, calculated by multiplying its distance to the centre by that cluster's scaling factor. More populous clusters receive larger penalties, encouraging assignments to smaller clusters. Update the centres and scaling factors repeatedly, smoothing weight changes to reduce oscillations.

For our model: Calculate cluster population by summing node population weights. This encourages balanced, geographically concentrated groups

Common limitation: These methods seek good partitions without guaranteeing the global optimum. Connectedness must be explicitly enforced or checked.

THE LANDSCAPE

FAMILY	WHAT IT DOES WELL	WHAT IT GIVES UP
Multi-kernel growth Vickrey (1961)	Fast; contiguity follows from how districts grow	Compactness only indirect
Location / allocation Hess et al. (1965)	Compactness, via minimum total inertia	No contiguity constraint — maps fixed by hand
● **Exact / set partitioning** Garfinkel & Nemhauser (1970)	Provably optimal; all criteria pre-checked	Does not scale past small instances
● **Local search & genetic** Bozkaya (2003), Forman & Yue (2003)	Explores an enormous solution space quickly	No guarantee; disconnected districts need clean-up
● **Graph & tree partitioning** Yamada (2009), DeFord et al. (2021)	Contiguity is structural, not repaired	Sampling distribution is not uniform

● the three we build on

THE DATA

SHRUG

devdatalab.org/shrug

GEOMETRY + POPULATION

- Open village and town-level polygons from the 2011 Census the contiguity graph is built directly from these.
- 500,000 villages, 8,000 towns, 4,000 legislative constituencies, spanning 25 years.
- Aggregated under both the 3rd and 4th delimitation boundaries

LOKDHABA

Trivedi Centre, Ashoka University

VOTES

- Cleaned Lok Sabha and Vidhan Sabha results from 1962 onward.
- Candidate and constituency-level records built from ECI sources, extended with contextual variables.

THE PLAN

1. **Synthetic grid**
   Small enough to enumerate every legal map and check both methods against it.
2. **One mid-sized state**
   All-India is too big of a challenge. Start by Solving for a Mid Size State
3. **Report a series of results**
   Not just a single most optimal output

https://lokdhaba.ashoka.edu.in/dash
https://devdatalab.org/shrug_download/